

A design-focused paper presenting the **architecture and development of the RAG-based QA system. It will detail the system components—document corpus creation, retrieval module, LLM integration, and interface—and explain how design decisions meet the ODL requirements. This corresponds to building and demonstrating the artefact.**

RAG-based QA systems integrate four key components (corpus creation, retrieval modules, LLM integration, and user interfaces) with hybrid retrieval and modular architectures to achieve operational requirements for scalability and precision.

Abstract

Twenty-four studies on retrieval-augmented generation (RAG) systems outline design choices for building question-answering artefacts. The papers describe architectures that integrate four key components:

1. Corpus creation—from custom datasets of 500+ pages to large public benchmarks (up to 24 million documents).
2. Retrieval modules employing sparse (e.g., BM25), dense, hybrid, or graph-based methods that enhance recall and precision.
3. LLM integration using fine-tuning, parameter-efficient strategies (LoRA, QLoRa, quantization), or multi-agent pipelines to boost answer relevance.
4. User interfaces ranging from web portals and APIs to modular dashboards that support deployment and reproducibility.

Several studies report quantitative benefits: F1 scores between 35 and 57, accuracy improvements up to 99% in domain-specific settings, and speedups of 10–100× in some implementations. Designs featuring hybrid retrieval (in 20 of the studies), modular and multi-agent architectures, and explicit relevance assessment meet intended operational design criteria for scalability, reliability, and contextual precision.

Paper search

Using your research question "A design-focused paper presenting the **architecture and development** of the RAG-based QA system. It will detail the system components—document corpus creation, retrieval module, LLM integration, and interface—and explain how design decisions meet the ODL requirements. This corresponds to building and demonstrating the artefact.", we searched across over 126 million academic papers from the Semantic Scholar corpus. We retrieved the 500 papers most relevant to the query.

Screening

We screened in sources that met these criteria:

- **Technical Architecture Description:** Does the study provide detailed technical architecture description of a RAG-based QA system, including specific LLM integration methods?
- **Implementation Details:** Does the paper include practical implementation details rather than purely theoretical discussion?

- **RAG Components:** Does the study describe both retrieval module implementation and document corpus management specifically for RAG systems?
- **System Integration:** Does the study demonstrate complete system integration rather than focusing on isolated components?
- **RAG Integration:** Does the study explicitly include RAG components in its QA system implementation?
- **Empirical Evidence:** Does the study provide technical evaluation results or implementation case study evidence?
- **System Comparison:** Does the study include comparison with other architectural approaches or clear justification for the chosen architecture?

We considered all screening questions together and made a holistic judgement about whether to screen in each paper.

Data extraction

We asked a large language model to extract each data column below from each paper. We gave the model the extraction instructions shown below for each column.

- **System Components:**

Identify and list the key architectural components of the RAG system. Look in the methods or system design sections for detailed descriptions. Include:

- Retrieval module details (e.g., embedding model, retrieval strategy)
- Language model used
- Any unique preprocessing or integration techniques
- Interface or deployment approach

If multiple components are described, list them all. If any component is not clearly specified, note "Not specified" and provide any available partial information.

- **Document Corpus Characteristics:**

Describe the document corpus used in the RAG system:

- Source of documents (e.g., custom dataset, existing datasets)
- Languages supported
- Total number of documents or size of corpus
- Document types (e.g., text, OCR-processed, multilingual)
- Any preprocessing or filtering applied to the corpus

If exact numbers are not provided, use ranges or qualitative descriptions. If information is incomplete, note which specific details are missing.

- **Unique Design Innovations:**

Identify and describe any novel design approaches or innovations in the RAG system:

- Specific techniques to address challenges (e.g., hallucination, retrieval accuracy)
- Unique fine-tuning or alignment methods
- Innovative integration of retrieval and generation components

Provide a clear, concise description of the innovation, including its purpose and potential impact. If no unique innovations are identified, state "No specific innovations reported".

- **Evaluation Metrics:**

List all performance metrics used to evaluate the RAG system:

- Specific metrics (e.g., F1 score, EM, BLEU)
- Datasets used for evaluation
- Performance results for each metric

Capture both quantitative metrics and any qualitative assessment. If multiple datasets were used, list results for each separately. If exact numbers are not clear, use ranges or descriptive terms.

- **Computational Resources:**

Describe the computational requirements:

- Models used (specific versions)
- Hardware requirements (GPU, CPU)
- Model size
- Any optimization techniques (quantization, pruning)

If multiple model configurations are discussed, capture details for each. If specific hardware specs are not provided, note the general computational complexity or resource needs.

Results

Characteristics of Included Studies

Study	Study Focus (Retrieval-Augmented Generation Component)	Architecture Type	Implementation Scale	Key Innovation	Full text retrieved
Saha et al., 2025	Retrieval, corpus, large language model (LLM) integration	Custom QuIM-RAG, inverted index	Custom corpus, 500+ pages	Question-to-question inverted index matching	No
Oubah and Şener, 2024	Multilingual retrieval, optical character recognition (OCR), LLM	Transformer-based, OCR, hybrid	Multilingual, 6,500+ samples	Multilingual semantic retrieval, OCR integration, fine-tuned E5	Yes

Study	Study Focus (Retrieval-Augmented Generation Component)	Architecture Type	Implementation Scale	Key Innovation	Full text retrieved
Pu et al., 2024	Domain-specific retrieval-augmented generation (RAG) (exploratory data analysis)	Custom RAG, contrastive learning	ORD-QA, 290 chunks	Contrastive learning, reranker, domain LLM	Yes
Liu et al., 2024	LLM-aligned retrieval	Two-stage fine-tuning, low-rank adaptation (LoRA)	Open-domain question answering, large datasets	Invariance loss, LoRA-based retriever	Yes
Dong et al., 2024	Preference alignment	DPA-RAG, multi-stage alignment	Wikipedia, 100+ documents	Dual preference alignment, query augmentation	Yes
Shi et al., 2024	Modular RAG, efficiency	Four-module synergy	6 question answering datasets	Query rewriting+, knowledge filter, memory reservoir	Yes
Wang et al., 2025	Financial, multi-modal	FinSage, multi-aspect	FinanceBench, real deployment	Multi-modal pipeline, direct preference optimization (DPO)	No
Liu et al., 2025	Multi-agent, multimodal	HM-RAG, hierarchical agents	ScienceQA, CrisisMMD	re-ranking Multi-agent, multi-source retrieval, expert refinement	Yes
Hou and Zhang, 2024	Dietary supplement question answering	RAG + knowledge graph	iDISK2.0, 174,000 entities	LLM-guided subgraph retrieval, user interface	Yes
Opoku et al., 2025	Domain-specific, graph	DO-RAG, hybrid graph/vector	Database, electrical domains	Agentic chain-of-thought, graph+vector fusion	No

Study	Study Focus (Retrieval-Augmented Generation Component)	Architecture Type	Implementation Scale	Key Innovation	Full text retrieved
Stuhlmann et al., 2025	Biomedical question answering	Hybrid BM25+MedCPT	PubMed, 24 million documents	Hybrid retrieval, scalable deployment	Yes
Cao et al., 2025	PDF question answering, neural-symbolic	NeuSym-RAG, hybrid	AIRQA-REAL, 6,797 PDFs	Multi-view chunking, schema parsing	Yes
Aitymbetov et al., 2024	Multi-document question answering, efficiency	Lightweight, two-step	StrategyQA	BM25+MixedBreadth	No
Hou et al., 2024	Dietary supplement question answering	RAG + knowledge graph	iDISK2.0, 174,000 entities	Biomedical knowledge graph + LLM, noise reduction	No
Zhao et al., 2025	Graph-based RAG	E ² GraphRAG, adaptive	No mention found	Bidirectional entity-chunk index, adaptive retrieval	No
Feng et al., 2024	Network operations question answering, efficiency	EasyRAG, dual-route	Chinese, 4 HTML packages	Dual sparse retrieval, LLM reranker, image filtering	Yes
Kim et al., 2025	Financial question answering, hybrid	Three-phase RAG	7 finance datasets	Hybrid dense+sparse, DPO, markdown restructuring	Yes
Zha, "SMARTFin-RAG"	Financial RAG evaluation	Modular, live-demo	Financial documents, live	Modular runtime, document-centric evaluation, LLM-as-judge	No

Study	Study Focus (Retrieval-Augmented Generation Component)	Architecture Type	Implementation Scale	Key Innovation	Full text retrieved
Wang et al., 2024a	Long document question answering, graph	GARLIC, hierarchical weighted directed acyclic graph (HWDAG)	LongBench, 4 datasets	LLM-guided graph search, attention-based retrieval	Yes
Visciarelli et al., 2024	Legal question answering, domain adaptation	RAG + quantized low-rank adaptation (QLoRa)	Emilia-Romagna law	QLoRa quantization, instruction tuning	No
Wang et al., 2024b	Relevance-aware RAG	REAR, rank head	Natural Questions, TriviaQA, SQuAD	Explicit relevance assessment, bi-granularity fusion	Yes
Vujić et al., 2025	Graph-based, modular	Dense+graph+rerank	No mention found	ChromaDB, PageRank, modular pipeline	No
Dong et al., "Talk Before You Retrieve"	Medical question answering, agentic	Discuss-RAG, plug-in	4 medical question answering datasets	Agent-led discussion, summarizer/decision agents	No
Finardi et al., 2024	RAG best practices	Multi-stage, hybrid	Harry Potter, Portuguese	Hybrid BM25+ADA-002, multi-stage rerank	Yes
Chen et al., 2025	Multi-agent reinforcement learning RAG	MMOA-RAG, multi-agent proximal policy optimization (MAPPO)	HotpotQA, 2Wiki, AmbigQA	Multi-agent reinforcement learning, joint module optimization	Yes

Architecture Types:

- Hybrid architectures (dense, sparse, or graph retrieval): 7 studies
- Graph-based or graph+vector architectures: 4 studies

- Knowledge graph integration:2 studies
- Multi-agent or agentic architectures:3 studies
- Modular architectures:3 studies
- Custom or uniquely named RAG architectures:11 studies
- Parameter-efficient fine-tuning (LoRA/QLoRa):2 studies
- Plug-in architecture:1 study
- Transformer-based architecture:1 study

Key Innovations:

- Retrieval enhancement (hybrid retrieval, reranking, query rewriting, chunking, etc.):20 studies
- Graph/vector fusion or graph search:3 studies
- Knowledge graph integration:2 studies
- Multi-agent or agentic approaches:3 studies
- Preference alignment:1 study
- Multi-modal retrieval:2 studies
- Domain adaptation:2 studies
- Modular pipelines:3 studies
- Evaluation/LLM-as-judge:1 study
- Efficiency/low-resource settings:1 study
- Multilingual retrieval:1 study

Implementation Scale:

- Large-scale datasets (thousands of documents/entities or real-world deployment):9 studies
- Domain-specific datasets (finance, law, medical, science):7 studies
- Real-world or live deployment:2 studies
- No mention found for implementation scale:2 studies
- Small or medium-scale datasets (hundreds of documents or chunks):3 studies

System Architecture Analysis

Core Retrieval-Augmented Generation Components

All included studies describe the core pipeline for retrieval-augmented generation systems, consisting of:

- Document corpus creation:Approaches range from custom datasets to large-scale public benchmarks.
- Retrieval module:Implementations include traditional sparse retrieval (such as BM25), dense embedding models, graph-based retrieval, and hybrid neural-symbolic methods.
- Large language model (LLM) integration:Both open-source (e.g., LLaMA, Qwen, DeepSeek) and commercial (e.g., GPT-4, Gemini) models are used, with several studies applying fine-tuning, quantization, or parameter-efficient adaptation (such as LoRA or QLoRa).
- Interface or deployment mechanism:Where described, interfaces include web portals, application programming interfaces (APIs), and modular dashboards.

Integration Patterns

Key integration patterns identified across studies:

- Multiple retrieval strategies: Many studies combine dense and sparse retrieval, graph and vector retrieval, or neural and symbolic methods to improve recall and precision.
- Multi-agent or multi-module architectures: Some studies (e.g., Liu et al., 2025; Cao et al., 2025; Opoku et al., 2025) use parallel or hierarchical processing of queries and evidence.
- Preference alignment and explicit relevance assessment: These are increasingly used to reduce hallucination and improve answer faithfulness.

Performance Optimization Approaches

Component Type	Design Pattern	Implementation Method	Performance Impact
Retrieval	Hybrid (dense+sparse, graph+vector)	BM25+MedCPT, ChromaDB+PageRank, multi-path	Improved recall/precision, faster retrieval
LLM	Parameter-efficient tuning	LoRA, QLoRa, quantization	Reduced memory, faster inference
Reranker	Cross-encoder, DPO, BGE	Fine-tuned rerankers, DPO, BGE	Higher answer relevance, compliance
Pipeline	Multi-agent, modular	MAPPO, plug-and-play, modular runtime	Joint optimization, scalability
Preprocessing	Multi-modal, chunking, OCR	Multi-view chunking, OCR, schema parsing	Broader coverage, improved context

Design Patterns and Performance Impacts:

- Hybrid retrieval (dense+sparse, graph+vector): Improved recall and precision, faster retrieval
- Parameter-efficient tuning for LLMs (LoRA, QLoRa, quantization): Reduced memory usage, faster inference
- Reranking (cross-encoder, DPO, BGE): Higher answer relevance, compliance
- Multi-agent and modular pipeline designs: Joint optimization, scalability
- Preprocessing (multi-modal, chunking, OCR): Broader coverage, improved context

We did not find mention of missing data for design pattern or performance impact in the studies summarized above. Some studies reported multiple design patterns or impacts per component.

Implementation Outcomes

Performance Metrics

Study	Metric Category	Measurement Approach	Results Range	Impact Factor
Saha et al., 2025	BERT-Score, RAGAS	Custom corpus	Outperforms baseline	Improved question answering accuracy

Study	Metric Category	Measurement Approach	Results Range	Impact Factor
Oubah and Şener, 2024	F1, Exact Match (EM), BLEU, RAGAS	Multilingual question answering	F1: 0.47–0.75	Multilingual, OCR
Pu et al., 2024	Recall, BLEU, ROUGE-L	ORD-QA, exploratory data analysis	Recall, BLEU improved	Domain-specific question answering
Liu et al., 2024	EM, Accuracy at 5/20	TriviaQA, Natural Questions, PopQA	EM: 53.6–75.3	LLM-aligned retrieval
Dong et al., 2024	Hit at 1, F1	Natural Questions, TriviaQA, HotpotQA, WebQSP	Outperforms baseline	Preference alignment
Shi et al., 2024	F1, Hit Rate, Recall	6 question answering datasets	F1: 35–57	Modular synergy
Wang et al., 2025	Recall, Accuracy	FinanceBench	Recall: 92.5%	Compliance question answering
Liu et al., 2025	Accuracy, ROUGE-L, BLEU	ScienceQA, CrisisMMD	Accuracy: 58–94%	Multi-agent, multimodal
Hou and Zhang, 2024	Accuracy	True/False, multiple choice	95–99%	Dietary supplement question answering, knowledge graph + LLM
Opoku et al., 2025	Recall, Relevancy	Domain question answering	Recall: near-perfect	Graph+vector fusion
Stuhlmann et al., 2025	Recall, Precision, F1	PubMed, BIOASQ	Recall: 0.57	Hybrid retrieval
Cao et al., 2025	LLM-based, subjective	AIRQA-REAL, M3SciQA	No mention found	Neural-symbolic, PDF
Aitymbetov et al., 2024	Accuracy, Evidence Quality	StrategyQA	Superior to baseline	Lightweight, efficient
Hou et al., 2024	Accuracy	True/False, multiple choice	95–99%	Biomedical knowledge graph + LLM
Zhao et al., 2025	Speed, question answering performance	No mention found	10–100x speedup	Graph-based, adaptive
Feng et al., 2024	Accuracy, Similarity	GLM4, Chinese	No mention found	Efficient, dual-route
Kim et al., 2025	Normalized Discounted Cumulative Gain at 10 (NDCG@10), Relevance	7 finance datasets	NDCG@10 improved	Hybrid, DPO

Study	Metric Category	Measurement Approach	Results Range	Impact Factor
Zha, "SMARTFinRAG"	Hit rate, Mean Reciprocal Rank (MRR), NDCG	Financial documents	No mention found	Modular, LLM-as-judge
Wang et al., 2024a	F1, ROUGE-L, BLEU-4	LongBench	Outperforms baseline	LLM-guided graph
Visciarelli et al., 2024	Perplexity, Expert Review	Legal question answering	No mention found	QLoRa, domain adaptation
Wang et al., 2024b	EM, F1, Hit at 1, Jaccard similarity	Natural Questions, TriviaQA, SQuAD	No mention found	Relevance-aware
Vujić et al., 2025	Precision, Recall	No mention found	No mention found	Modular, graph-based
Dong et al., "Talk Before You Retrieve"	Answer accuracy	BioASQ, PubMedQA	+12–17%	Agentic, medical question answering
Finardi et al., 2024	MRR at 10, Recall	Harry Potter, Portuguese	MRR@10: +35%	Hybrid, best practices
Chen et al., 2025	Accuracy, EM, F1	HotpotQA, 2Wiki, AmbigQA	Accuracy: 38–42%	Multi-agent reinforcement learning

Metric Categories:

- Most common metrics: Accuracy (7 studies), F1 (7 studies), Recall (7 studies)
- Other metrics: Exact Match (4 studies), BLEU and ROUGE-L (3 studies each), Hit at 1 (2 studies), Precision (2 studies), NDCG/NDCG@10 (2 studies), MRR/MRR@10 (2 studies), Relevancy/Relevance (2 studies), and a variety of single-use metrics (BERT-Score, Accuracy at 5/20, Hit Rate, Evidence Quality, Speed, Similarity, BLEU-4, Jaccard similarity, Perplexity, Expert Review, LLM-based, Subjective, Answer accuracy)
- Multiple metrics: Most studies reported more than one metric

Measurement Approaches:

- Standard question answering datasets (e.g., Natural Questions, TriviaQA, SQuAD, HotpotQA): 9 studies
- Domain-specific datasets (finance, biomedical, legal, Harry Potter, and others): 11 studies
- Multilingual question answering: 1 study
- Multimodal approach: 1 study
- Custom corpus: 1 study
- No mention found for measurement approach: 2 studies

Results Range:

- Quantitative results (accuracy, F1, recall, or other metrics with numerical values or ranges): 13 studies
- Qualitative results ("outperforms baseline", "improved", or "superior to baseline"): 6 studies
- No mention found for detailed results: 6 studies

System Reliability

- Technical reliability: Most studies assess reliability using technical metrics such as accuracy, recall, and precision.
- Scalability and efficiency: Some studies (e.g., Stuhlmann et al., 2025; Wang et al., 2025) report on scalability and efficiency.
- User-facing reliability: We did not find mention of user-facing reliability (such as uptime or error rates) in the available full texts or abstracts.

User Interface Effectiveness

- Reporting: Only a minority of studies describe user-facing portals, dashboards, or APIs.
 - Evaluation: Where evaluated, interfaces are described as user-friendly and efficient, but we did not find mention of systematic user studies or usability metrics.
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Thematic Analysis

Document Processing and Retrieval Strategies

- Corpus creation: Highly domain-dependent, with some studies constructing custom datasets and others using large-scale public benchmarks.
- Preprocessing techniques: Include chunking (by section, token, or event), OCR for scanned or PDF documents, and knowledge graph construction for structured domains.
- Retrieval strategies: Increasingly hybrid, combining dense and sparse methods, or integrating symbolic and neural approaches.
- Emerging trends: Graph-based retrieval and multi-agent decomposition are used for complex or multi-modal corpora.

LLM Integration Techniques

- Model types: Both open-source (LLaMA, Qwen, DeepSeek) and commercial (GPT-4, Gemini) models are used.
- Adaptation methods: Fine-tuning, LoRA, QLoRA, and DPO are common; quantization (e.g., 4-bit) is used for deployment efficiency.
- Multi-agent systems: Some studies treat the RAG pipeline as a multi-agent system, enabling joint optimization of retrieval, reranking, and generation.
- Faithfulness: Preference alignment and explicit relevance assessment are used to improve answer faithfulness and reduce hallucination.

Interface Design Patterns

- Types: Web-based portals, modular dashboards, and plug-and-play modules are described in some studies.
- Deployment: Some systems are deployed in real-world settings (e.g., FinSage in online financial meetings), while others provide APIs or open-source code for reproducibility.
- Evaluation: We did not find mention of systematic user studies or usability metrics.

System Scalability Solutions

- Architectures: Modular architectures, memory-optimized parallelization, and parameter-efficient LLM adaptation are used to address scalability.
- Deployment: Some studies report deployment on multi-GPU clusters, while others emphasize lightweight or efficient models for low-resource settings.
- Reproducibility: Open-source code and benchmarks are increasingly provided to facilitate reproducibility and scalability.

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